FINANCIAL® DATA EXCHANGE

Consensus Standard Data Format

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FDX Traffic Light Protocol (TLP) Green:

Recipients may share TLP GREEN information with peers, trusted government and critical infrastructure partner organizations, and service providers with whom they have a contractual relationship, who have a need-to-know but not via publicly accessible channels.

Key Words

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" when used in ALL CAPS in this document are to be interpreted as described in IETF RFC Editor BCP 14 (with IETF RFC 2119 "Key words for use in RFCs to Indicate Requirement Levels" and IETF RFC 8174 "Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words").

Revision History

Document Version	Notes	Date
1.0	Initial Document Release from RFC 0352	**May 2025**

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1 Overview: Consensus Standard Data Format (CSDF)

This document describes the Financial Data Exchange's "consensus standard data format" (CSDF).

The Consumer Financial Protection Bureau's (CFPB) Personal Financial Data Rights Rulemaking (PFDR Rulemaking) describes obligations for eligible U.S. Data Provider firms to maintain developer interfaces that enable the sharing of "covered data in a standardized and machine-readable format." The PFDR Rulemaking further states: "indicia that the format satisfies this requirement includes that it conforms to a consensus standard." A consensus standard is defined to mean a standard that is adopted by a recognized standard setter.

The Financial Data Exchange (FDX) is a CFPB-recognized standard-setter. This document defines the specific requirements a Data Provider's developer interface must meet to conform with the FDX consensus standard data format, as may be relevant for achieving an indicia of compliance.

For more information about FDX and its standard-setting process, visit financialdataexchange.org.

The role of FDX CSDF alongside the broader FDX API Specification

FDX has iteratively defined and published an array of technical standards, best practices, and guidelines. Together, these are collectively labeled as the FDX Application Programming Interface Specification (i.e., the "full" FDX API Spec). The most recent versions of the full FDX API spec are available here.

The full FDX API Spec covers a wide array of topics, account types, and data elements, some of which extend beyond "covered data" and "format" as defined in the PFDR Rulemaking.

As such, conformance to the full FDX API Spec by a Data Provider is not required to conform with the FDX CSDF. The CSDF captures a subset of the broader FDX API Spec. Organizations are encouraged to review the full FDX API spec. Organizations may choose to implement additional elements of the broader FDX API spec to enable efficient, secure, and interoperable data sharing.

FDX intends to maintain both the full FDX API Specification and the CSDF. FDX will adopt new versions of both over time, including to address evolving market needs and any changes in regulation. FDX expects that the CSDF will remain a subset of the broader FDX API specification.

2 Requirements of the CSDF

To conform with the FDX Consensus Standard Data Format (CSDF), a Data Provider's developer interface MUST meet the requirements set out in this section.

2.1 Requirement 1: API Schema Alignment and Version Requirements

A Data Provider MUST implement the **Applicable Resources** (see details below) from the following eligible FDX API specification versions (as defined within the API files):

- Core API (version 5.4 or 6.4)
- Customers API (version 5.4 or 6.4)
- Money-Movement API (version 5.4 or 6.4)
- Meta API(version 5.4 or 6.4)

API schema alignment includes an obligation to adhere to the following aspects of data formatting:

- **Endpoints & Methods**: URL path patterns that define where specific API operations can be performed; HTTP methods that can be used with endpoints (e.g., GET, POST).
- **Request Orchestration & Parameters**: order-of-operations guidance that defines request chaining; inputs required to make data requests, including query parameters, headers, body content, and API authorization details.
- **Response Format**: structure of the data that the API returns, including status codes, headers, body content (e.g., JSON format), error responses, and pagination.
- **Data Structures**: data entity definitions including types (e.g., string, integer), enums, and usage in API requests & responses.
- Certain additional requirements including transport methods and API credentials (e.g., API keys, access tokens) for authorizing requests.

These four API schema files—Core, Customer, Money-Movement, and Meta—leverage a shared Components YAML file for common schema definitions. Instead of repeating the common definitions in each API YAML file, they reference the corresponding definitions within the Components YAML. As necessary to implement the **Applicable Resources**, a Data Provider MUST also follow the formatting described in the Components YAML.

2.1.1 Applicable Resources

To be CSDF conformant, a Data Provider is NOT required to share all elements described in the Core API, Customers API, Money-Movement API, and Meta API.

Rather, for whichever data elements from the **Reference List** (described below) that a Data Provider **does** share, all of those provided elements MUST be shared in alignment with the technical formatting requirements described above.

Note: The **Reference List** is included in the FDX API specification (refer to the accompanying file "*CSDF Reference List.xlsx*" in the **Consensus Standard Data Format** folder).

The **Reference List** for CSDF is made up of the following data elements:

- All data elements in the FDX spec associated with the following account types:
  - Regulation E accounts
  - Regulation Z credit card accounts
  - Digital Wallet (or any other consumer financial product or service that facilitates of payments from a Reg E account or Reg Z credit card, excluding products or services that merely facilitate first party payments)
- Additional "technical data elements" that are not account information, but which help to facilitate sharing of the aforementioned Reg E, Reg Z, and Wallet account data (e.g., unique identifiers, error codes).

For clarity, a Data Provider is not obligated to return all of the data elements in the **Reference List** to be CSDF conformant.

- A Data Provider may share the subset of the **Reference List** that the Data Provider deems to be the "covered data" it is obligated to share under the PFDR Rulemaking.*
- Alternatively, a Data Provider may share all of **Reference List** elements, as well as other elements defined in the full FDX API spec.

** FDX has not sought to define a consensus view on which specific data elements are "covered data" under the PFDR Rulemaking. Because the PFDR Rulemaking at times defines "covered data" in high-level terms, different parties may reasonably reach different interpretations of whether a specific data element is "covered data" that must be shared.*

The **Applicable Resources** are those resources (e.g., endpoint, schemas, enums) that are necessary to support sharing the data elements from the Reference List the Data Provider chooses to share.

FDX has tagged all Reference List fields in the YAML files. A separate instruction guide provides detailed steps on how to filter and view these tags. This guide is included in the FDX API specification (refer to accompanying file "*FDX CSDF Tagging Guide*" in the **Consensus Standard Data Format** folder).

Clarification regarding optionality:

Certain data elements may be accessible through multiple endpoints in the YAML files. If a Data Provider chooses to share a given data element from the Reference List, the Data Provider MUST implement at least one of the applicable endpoints for that element. Implementing additional endpoints to share the same data element is allowed but not required. For example, "Terms and Conditions" data (refer to *fdxapi.core.yaml*) may be returned as a machine-readable PDF, structured fields, or both, if so defined in the YAML.

2.2 Requirement 2: API Security

a. A Data Provider's developer interface MUST implement an approved security profile as defined in the *FDX Consensus Standard Data Format v1 API Security Profile* section of the *FDX Consensus Standard Data Format (v1) API Security Model* document. Specifically, a Data Provider MAY choose either of the following security profiles for the relevant endpoints (as defined in the *FDX Consensus Standard Data Format v1 API Security Profile* section of the *FDX Consensus Standard Data Format (v1) API Security Model* document), but MUST fully implement the requirements of the chosen profile:
    i. **FDX Consensus Standard Data Format v1 API Security Profile – Green** (based on OAuth 2.0)
    ii. **FDX Consensus Standard Data Format v1 API Security Profile – Blue** (based on FAPI 2.0)
b. Conditional Requirements: The following requirements apply if a Data Provider implements the specified functionality:
    i. **FDX Message Encryption**: If agreements between implementers specify message-level encryption (e.g., encrypted at the application level), the implementations MUST conform with the specification set forth in the *FDX Consensus Standard Data Format (v1) API Security Model*.
    ii. **FDX Step-up Authentication**: If agreements between implementers specify step-up authentication, the implementations MUST conform with the specifications set forth in the *Consensus Standard Data Format* section of the *FDX API Security Model* document.

Implementers should refer to the *Consensus Standard Data Format* section of the *FDX API Security Model* document for the complete normative text, including flows, token handling, message encryption mechanics, and any scope checks. The requirements in the *Consensus Standard Data Format* section of the *FDX API Security Model* document supersede any conflicting descriptions found in other FDX documents.

> **Note**: FDX will continue to assess different communication protocol requirements, including regarding token security, encryption methods, and methods for establishing trust between entities. FDX may evolve these requirements as part of subsequent versions of the Consensus Standard Data Format.

3 Consent Management

Secure, user-permissioned data sharing requires different parties (including Data Providers, data access platforms, and data recipients) to exchange information about what the consumer has consented to share.

Standardized methods for consent management may address such topics as (but not limited to): how data clusters and scopes are defined; how parties exchange information about what the consumer consents to share; how data recipients are registered with Data Providers; methods for consent revocation and reauthorization; and transmission of signed authorization disclosures.

FDX is not including requirements covering such consent management methods as part of this first version of the CSDF. FDX has chartered a group to work on defining and refining standards for consent management that FDX may include in a future version of the CSDF.

4 Validating Conformance with the CSDF

FDX will offer a conformance test to determine if a Data Provider is aligned with the requirements of the CSDF. Data Providers will have the option to engage FDX to validate their developer interface using this test.

- That CSDF conformance test will provide visibility into which of the Reference List data fields were shared in the test and if the fields shared were correctly formatted.
- If all of the fields that the Data Provider elects to share for testing are formatted correctly, and if the Data Provider meets all of the other requirements of the CSDF, then the Data Provider will be deemed by FDX to be conformant with the CSDF for those fields.
- If those requirements aren't met, the Data Provider will not receive a passing conformance result unless or until the issues are resolved and the test is re-run with correct formatting.
- Upon successful completion of the test (with all shared fields being correctly formatted), FDX will make visible both to that Data Provider and to FDX members: (1) which fields the Data Provider submitted for conformance testing and (2) that the Data Provider has passed.
- The conformance test result will reflect the Data Provider's implementation as tested on a specific date. FDX will offer Data Providers an opportunity to undergo testing again at a subsequent date (e.g., to validate conformance if a Data Provider later updates its API developer interface).
- For clarity: FDX's assessment of a Data Provider's conformance with the CSDF will not be based on which data elements the Data Provider does or doesn't make available, but instead whether the correct format is used for any elements that are shared.

  Note: FDX may offer additional conformance validation tools that allow organizations to validate their use of different data scopes and use cases beyond the CSDF and unrelated to the PFDR Rulemaking.

5 Evolution of the FDX CSDF and Future Versions

FDX intends to publish subsequent versions of the CSDF. As FDX members reach consensus and define common, interoperable methods to address additional aspects of data format, these may be included in future versions. FDX may also address in subsequent CSDF versions any modified or expanded requirements to support changes in regulation.

As subsequent versions of the CSDF are released, FDX intends to establish a timeline for retiring older versions (i.e., requiring Data Providers to conform to updated versions to still be deemed conformant by FDX with the CSDF). FDX intends to maintain and provide CSDF conformance testing tools for every supported version of the CSDF. Specific dates for retiring old versions will be set in the future through a similar consensus process by FDX, considering factors including interoperability, reasonable notice and upgrade timelines, and market needs.

CSDF Spec and Reference List Version	CSDF Conformance Target Version	Date of publication	Date retired	Associated version of FDX API Spec
Version 1	v5.4	**April 2025 (TBC)**	**TBD**	**FDX API Spec v5.4**
Version 1	v6.4	April 2025 (TBC)	TBD	FDX API Spec v6.4
Version 2	vX.X	TBD	TBD	TBD

Definitions

Data Provider (DP): An entity that (1) controls or possesses information concerning a financial product or service that the customer obtained from that entity and that (2) the customer is directing to share such information with another party. For example, this may include a company that provides a Regulation E, Regulation Z, or Wallet account to a consumer.